

Sharpness-Consistent Cross-Domain Recommendation for Cold-Start Items

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Abstract

Cold-start remains a fundamental challenge in recommendation systems due to the scarcity of interaction data. Recent methods address this issue by leveraging semantic ID embeddings and cross-domain transfer techniques, achieving notable progress. However, the common practice of learning semantic ID embeddings and training the recommendation model in separate stages hinders the generalization capability of semantic IDs throughout the training process. In this work, we propose **Sharpness-Consistent Cross-Domain Recommendation (SC²Rec)**, a novel framework designed to enhance the generalization of semantic ID-based models in cold-start scenarios. SC²Rec alternately optimizes the sharpness of the loss landscape and enforces landscape consistency between warm and cold domains, leading to unified and flatter minima and improved generalization. Extensive experiments on industrial datasets demonstrate the effectiveness of SC²Rec. Furthermore, we release a high-quality dataset to facilitate further research in this area.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Recommender Systems; Cold-Start; Cross-Domain Recommendation

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1 Introduction

The ever-expanding digital landscape has made Recommender Systems (RS) central to guiding users toward relevant content. A persistent challenge, especially for industrial deployment, is the *cold-start item problem*—where newly introduced items lack sufficient interaction data, hampering effective personalization [37]. Cross-domain recommendation (CDR) offers a potential remedy by transferring knowledge from data-rich (warm) domains to data-sparse (cold) domains [3, 36]. Classical CDR approaches have explored feature- or embedding-level transfer, notably through adversarial alignment methods [28, 29]. These techniques align feature distributions between domains, thereby enhancing the transferability of both sparse (e.g., user or item attributes) and dense (e.g., semantic embeddings) features. However, while distributional alignment can reduce domain discrepancies, it fails to transfer collaborative filtering signals, as traditional ID embeddings are unique and inherently non-transferable.

Recently, large language models (LLMs) and multimodal encoders have been leveraged to distill items into semantic ID codes via vector quantization [10, 26]. These semantic ID codes preserve both collaborative memorization ability and representational transferability, leading to substantial improvements in cold item recommendation.

Despite these advances, prevailing industrial RS systems typically adopt decoupled pipelines: semantic ID encoders are trained independently of real-time ranking modules for scalability, resulting in increasing misalignment between semantic ID features and evolving user behavior patterns. SaviorRec [33] partially addresses this gap by making codebooks learnable within the ranking model, yet continual adaptation risks overfitting to the dominant (warm) domain, diminishing generalization to cold items—as confirmed by pronounced sharp minima in their loss landscape (Figure 1).

Loss sharpness has emerged as a crucial factor in model generalization: sharp optima indicate parameter sensitivity to small perturbations, often linked to poor transferability [6, 9]. While sharpness-aware minimization (SAM) [7, 35] can regularize towards flatter minima, naïvely combining warm and cold losses is suboptimal; warm-domain data dominates sharpness regularization, leaving cold-domain parameters insufficiently regularized. Our key

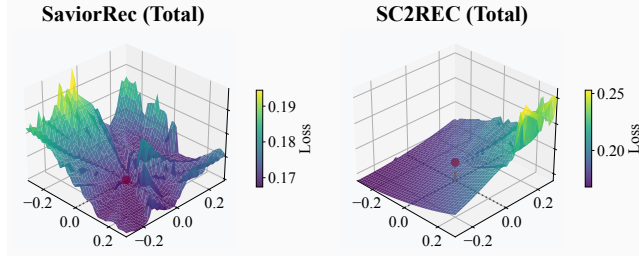


Figure 1: Loss landscape visualization. We adopt the visualization technique from Li et al. [16], which perturbs model parameters along two randomly selected directions to generate a three-dimensional loss surface.

insight is that effective cross-domain transfer not only requires flat minima for each domain, but also *consistency* in loss landscape geometry across domains for shared parameters—otherwise, the model may occupy a region that generalizes well in one domain but remains brittle in the other.

To overcome this limitation, we propose **Sharpness-Consistent Cross-Domain Recommendation (SC²Rec)**, a unified framework that alleviates domain overfitting by explicitly promoting flat, consistent minima in the loss landscape for both warm and cold domains. Specifically, SC²Rec applies sharpness-aware minimization to key transferable parameters within the cross-domain model and introduces a refiner network that directly enforces landscape flatness in both domains, fostering balanced generalization and improved cross-domain transfer. Our main contributions are summarized as follows:

- We propose **SC²Rec**, a framework that jointly optimizes for flat and consistent loss landscapes across domains to tackle cold-start recommendation.
- We release a novel benchmark dataset with over 500 real-world features and semantic IDs.
- Extensive experiments validate our method, showing significant AUC improvements of 1.51% and 0.61% on industrial datasets.

2 Related Work

2.1 Cross-Domain Recommendation

Cross-domain recommendation (CDR) mitigates data sparsity by transferring knowledge from a data-rich auxiliary domain. Early models like CoNet [11] and SARNET [24] rely on overlapping users or items to facilitate knowledge transfer through shared parameters. However, their effectiveness is limited in practical cold-start scenarios where such overlap is scarce [40].

To address limited overlap, feature-based alignment methods such as RecSys-DAN [29] and AA [28] employ adversarial learning to align feature distributions across domains. While promoting domain invariance, these methods often underutilize personalized ID information, potentially harming recommendation accuracy.

The rise of LLMs and multimodal learning has shifted focus to semantic item representations. MoRec [34] and Recformer [17] create universal encodings from multimodal or textual data, improving

generalizability at the risk of homogenized recommendations. Subsequent works like VQ-Rec [10] introduce discrete semantic IDs to better balance generalization with the memorization of collaborative patterns.

In practice, the decoupling of embedding and ranking pipelines for scalability introduces temporal misalignment, degrading cold-start performance. SaviorRec [33] addresses this with a learnable codebook but risks overfitting to the warm domain over time.

In summary, prior work highlights a core trade-off: balancing generalization for cold items with strong personalized memorization. This underscores the need for frameworks that achieve robust feature alignment while adapting to dynamic user behavior.

2.2 Sharpness-aware Minimization (SAM) in Recommendation

Recent advances in domain generalization emphasize the impact of the loss landscape’s geometry—specifically, solution sharpness—on model generalization [4, 6, 9]. Sharpness-Aware Minimization (SAM) introduces an explicit penalty for loss sharpness during optimization, guiding models toward flatter minima associated with improved out-of-distribution robustness [7].

Subsequent research has investigated SAM’s theoretical foundations, analyzing generalization bounds, convergence [22, 31], and computational strategies [5, 19, 21]. In recommendation contexts, SAM has been applied to sequential models [14], graph-based approaches [2], and cold user recommendation [35]. Most relevant work SCDR [35] employs SAM to user representations in matrix factorization frameworks [13]. However, these approaches focus on intra-domain scenarios and do not fully address sharpness discrepancies between domains—particularly in deep neural networks utilizing semantic IDs.

In our work, we pioneer the integration of sharpness-aware minimization with deep neural ranking models for semantic cold item recommendation. By enforcing sharpness consistency across warm and cold domains, our framework continually optimizes transferable parameters and semantic ID representations. This directly addresses the generalization decline induced by ongoing adaptation to behavior-rich warm domain data, establishing a principled pathway for robust cross-domain transfer.

3 Methodology

3.1 Problem Definition

The cold-start item problem in recommender systems refers to the challenge of making accurate recommendations for items (or users) with limited or no historical interaction data. Formally, let $x_{u,i}$ denote the concatenation of sparse and dense feature embeddings representing a user-item pair. We define the *source domain* $\mathcal{D}_{\text{warm}}$ as the collection of abundant user-item interactions, while the *target domain* $\mathcal{D}_{\text{cold}}$ consists of new items with scarce interaction data. In our business scenario, items with fewer than 30 page views (*pv*) are considered cold items. Our objective is to leverage the abundant data in the warm domain to enhance Click-Through Rate (CTR) estimation accuracy for cold items. Let $\mathcal{D} = \mathcal{D}_{\text{warm}} \cup \mathcal{D}_{\text{cold}}$ denote the entire training set, and $|\mathcal{D}|$ its size.

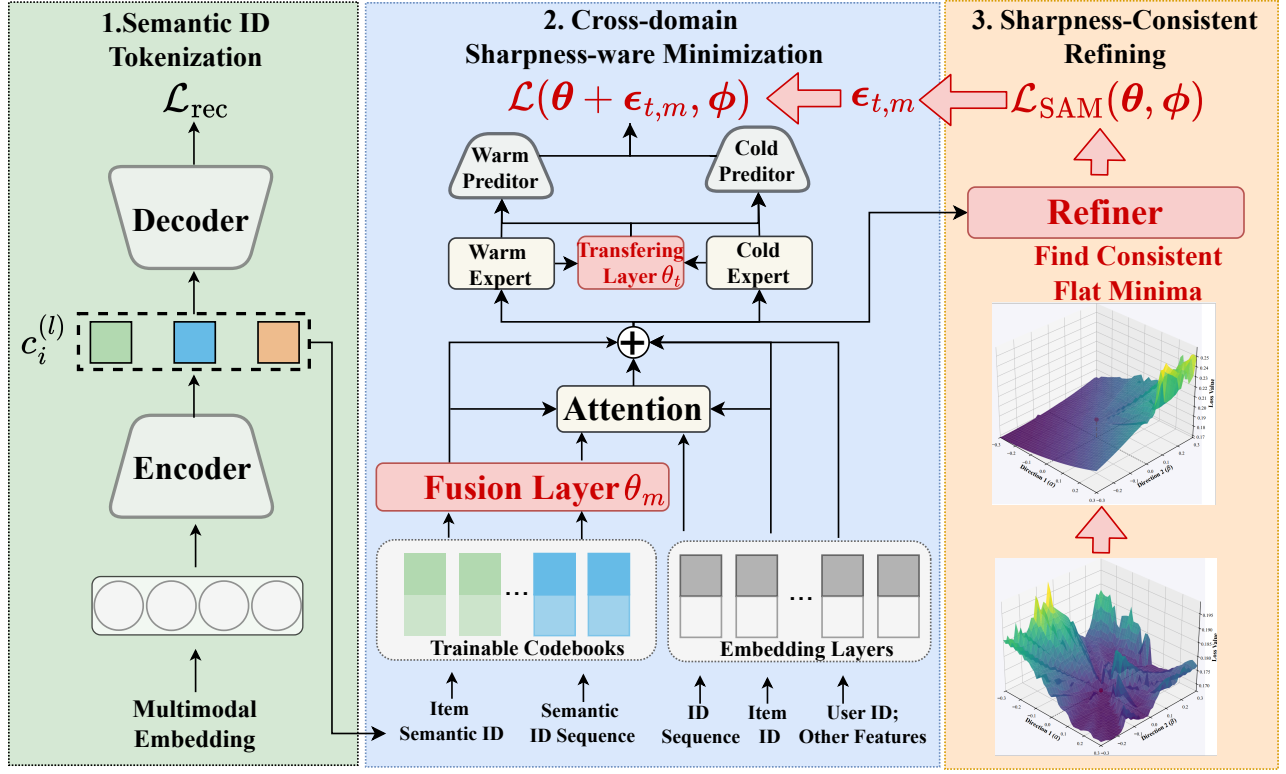


Figure 2: Overview of the SC²Rec framework. SC²Rec comprises three primary components: (1) Semantic ID Tokenization, (2) Cross-domain Sharpness-Aware Minimization, and (3) Sharpness-Consistent Refining. The symbol $\oplus$ denotes feature concatenation.

3.2 Semantic ID Tokenization

Traditional recommender systems rely on unique ID-based embeddings, which encode collaborative signals from user-item interactions but lack the ability to represent semantic content. Let e_u and e_i denote the embedding vectors for user u and item i , respectively; for a user u with an interaction history $\mathcal{H}_u = (i_1, \dots, i_t)$, their historic item embeddings are $(e_{i_1}, \dots, e_{i_t})$.

To address the semantic gap, we employ a residual quantization technique, specifically the Residual Quantized Variational Autoencoder (RQ-VAE) [23], which discretizes continuous item representations $\mathbf{z}_i$ obtained from multimodal large language models (LLMs) into discrete semantic codes using L codebooks $\{C_1, \dots, C_L\}$, with $C_l \in \mathbb{R}^{K_l \times d}$ and K_l increasing with l . The quantization proceeds iteratively:

$$\begin{aligned} \mathbf{r}_i^{(1)} &= \text{Enc}(\mathbf{z}_i), \\ c_i^{(l)} &= \arg \min_{k \in \{1, \dots, K_l\}} \left\| \mathbf{r}_i^{(l)} - C_l[k] \right\|_2^2, \quad l = 1, \dots, L, \\ \mathbf{r}_i^{(l+1)} &= \mathbf{r}_i^{(l)} - C_l[c_i^{(l)}]. \end{aligned} \quad (1)$$

The resulting sequence of code indices $\mathbf{c}_i = (c_i^{(1)}, \dots, c_i^{(L)})$ defines the discrete *semantic ID* for item i . The quantized representation $\mathbf{q}_i = \sum_{l=1}^L C_l[c_i^{(l)}]$ is passed through a decoder to reconstruct the original representation: $\hat{\mathbf{z}}_i = \text{Dec}(\mathbf{q}_i)$.

We train the model via a reconstruction loss:

$$\mathcal{L}_{\text{rec}} = \frac{1}{N} \sum_{i=1}^N \left\| \hat{\mathbf{z}}_i - \mathbf{z}_i \right\|_2^2, \quad (2)$$

where N is the number of items.

3.3 Cross-domain Sharpness Minimization

Most industrial recommender systems employ a decoupled architecture for computational efficiency, in which semantic ID embeddings and ranking models are trained independently. Recent approaches seek to alleviate the gap between multimodal item features and dynamically evolving user behavior patterns by learning item semantic IDs through randomly initialized codebooks within the ranking model [33]. However, these methods frequently absorb excessive behavior-specific knowledge from the warm domain, thus impairing the generalization and transferability of semantic IDs to cold domains.

To address this challenge, we propose a Refined Sharpness-Aware Minimization (SAM) framework for cross-domain recommendation. Our method aims to preserve strong generalization of semantic ID representations, ensuring robust performance in cold-start scenarios and facilitating knowledge transfer across domains. The technical details are as follows.

Table 1: Experimental results are presented on both industrial datasets (1% released) and industrial datasets (full). We report the CTR AUC and LogLoss (LL), averaged over five runs with different random seeds. Bold indicates the best result, and underlined indicates the second-best. “*” denotes a statistically significant improvement over the second-best ($p < 0.05$, two-sided t-test).

Method	Industrial (1%)				Industrial (Full)			
	Warm AUC	Cold AUC	Warm LL	Cold LL	Warm AUC	Cold AUC	Warm LL	Cold LL
Shbot [1]	0.6787	0.6946	0.1631	0.1234	0.7942	0.8193	0.1719	0.1332
MMOE [20]	0.6765	0.6929	0.1627	0.1239	0.7946	0.8195	0.1716	0.1330
STAR [25]	0.6788	0.6906	<u>0.1626</u>	0.1240	0.7938	0.8156	0.1721	0.1336
SARNet [24]	0.6789	0.6938	0.1631	0.1235	0.7939	0.8188	0.1721	0.1327
Hamur [18]	0.6792	0.6957	0.1627	0.1224	0.7942	0.8207	0.1717	0.1314
DAA [27]	0.6771	0.6627	0.1635	0.1274	0.7804	0.8028	0.1767	0.1364
Diff-MSR [30]	0.6769	0.6762	0.1636	0.1251	0.7919	0.8012	0.1726	0.1351
SaviorRec [33]	0.6751	0.6948	0.1627	0.1226	0.7901	0.8198	0.1725	0.1316
SCDR [35]	<u>0.6794</u>	<u>0.6999</u>	0.1628	<u>0.1220</u>	<u>0.7944</u>	<u>0.8249</u>	<u>0.1715</u>	<u>0.1310</u>
SC²Rec	0.6807*	0.7105*	0.1621*	0.1214*	0.7949*	0.8299*	0.1711*	0.1304*

3.3.1 Multi-Level Codebook Item Embedding. Traditional ID embeddings often lack semantic structure and generalization capacity, especially under cold-start conditions. Inspired by [33], we represent each item i using L residual quantization (RQ) layers. Each layer $l = 1, \dots, L$ is associated with a codebook $C_l \in \mathbb{R}^{K_l \times d}$, where K_l is the codebook size and d is the embedding dimension. The RQ code for item i at layer l is indexed by $c_i^{(l)} \in \{1, \dots, K_l\}$, and the codebook embedding for each item is obtained via a simple lookup:

$$\mathbf{v}_i^{(l)} = C_l[c_i^{(l)}] \in \mathbb{R}^d. \quad (3)$$

Concatenating all layers yields a composite embedding:

$$\mathbf{v}_i = \text{concat}(\mathbf{v}_i^{(1)}, \dots, \mathbf{v}_i^{(L)}) \in \mathbb{R}^{Ld}. \quad (4)$$

Distinct from the conventional summation of RQ code embeddings [26], we employ a fusion layer $f(\cdot; \theta_m)$ with parameters θ_m to synthesize the final semantic representation:

$$\mathbf{s}_i = f(\mathbf{v}_i; \theta_m) \in \mathbb{R}^d. \quad (5)$$

We optimize this RQ-based semantic embedding via sharpness-aware minimization (SAM), thereby enhancing representation generalization.

3.3.2 Attentive User Interest Modeling. Given a user u with interaction history $\mathcal{H}_u = (i_1, i_2, \dots, i_t)$, we construct a target-aware user representation for candidate item i by aggregating historical item embeddings via an attention mechanism [39]. Let $\mathbf{e}_i$ (ID embedding) or $\mathbf{s}_i$ (semantic ID embedding) denote the candidate item’s embedding, and $\mathbf{e}_{i_j}$ or $\mathbf{s}_{i_j}$ the historical item embeddings. To unify notation, let $\mathbf{g}_i$ represent the candidate embedding and $\mathbf{g}_{i_j}$ each historical item, where $\mathbf{g} \in \{\mathbf{e}, \mathbf{s}\}$. The attentive user interest is

computed as:

$$\mathbf{q}_i = \mathbf{g}_i \mathbf{W}^q, \quad \mathbf{k}_{i_j} = \mathbf{g}_{i_j} \mathbf{W}^k, \quad \mathbf{v}_{i_j} = \mathbf{g}_{i_j} \mathbf{W}^v, \quad (6)$$

$$\alpha_{ij} = \frac{\exp(\mathbf{q}_i \cdot \mathbf{k}_{i_j}^\top / \sqrt{d})}{\sum_{j'=1}^t \exp(\mathbf{q}_i \cdot \mathbf{k}_{i_{j'}}^\top / \sqrt{d})}, \quad (7)$$

$$\mathbf{z}_u = \sum_{j=1}^t \alpha_{ij} \mathbf{v}_{i_j}, \quad (8)$$

where $\mathbf{W}^q, \mathbf{W}^k, \mathbf{W}^v \in \mathbb{R}^{d_g \times d}$ are learnable projection matrices, d_g is the dimension of the selected embedding, and d is the attention dimension.

3.3.3 Cross-domain Transferring Networks. All features are concatenated as model input:

$$\mathbf{h}_{u,i} = \text{concat}(\mathbf{x}_{u,i}, \mathbf{e}_u, \mathbf{e}_i, \mathbf{s}_i, \mathbf{z}_u^{\text{sem}}, \mathbf{z}_u^{\text{ID}}), \quad (9)$$

where $\mathbf{z}_u^{\text{sem}}$ and $\mathbf{z}_u^{\text{ID}}$ are the attentive user interests based on semantic ID and ID streams, respectively.

To further decouple cold and warm-domain knowledge, we employ two expert networks with parameters θ_c (cold) and θ_w (warm):

$$\mathbf{o}_{\text{cold}} = f(\mathbf{h}_{u,i}; \theta_c), \quad (10)$$

$$\mathbf{o}_{\text{warm}} = f(\mathbf{h}_{u,i}; \theta_w). \quad (11)$$

Knowledge transfer is facilitated by cross-domain transferring networks parameterized by θ_t :

$$\hat{\mathbf{y}}_{ui}^{\text{warm}} = T_{\text{warm}}(\mathbf{o}_{\text{warm}} + f(\mathbf{o}_{\text{cold}}; \theta_t)), \quad (12)$$

$$\hat{\mathbf{y}}_{ui}^{\text{cold}} = T_{\text{cold}}(\mathbf{o}_{\text{cold}} + f(\mathbf{o}_{\text{warm}}; \theta_t)), \quad (13)$$

where T_{warm} and T_{cold} are domain-specific predictors.

3.3.4 Refined Sharpness-Aware Minimization. Standard optimization of the training loss may lead semantic ID embeddings to overfit

user behaviors from the warm domain, thereby reducing generalization. Inspired by [15], we adopt refined sharpness-aware minimization to regularize training and encourage robust generalization. Empirical studies in domain generalization have shown that solutions corresponding to flatter minima in the loss landscape—measured by sharpness—tend to generalize better [4, 6, 9].

Formally, for trainable parameters θ , SAM solves the min-max objective:

$$\min_{\theta} \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}(\theta + \epsilon), \quad (14)$$

where $\mathcal{L}$ is the loss function, and ϵ is a perturbation bounded by norm ρ . The inner maximization finds the most adversarial parameter perturbation, while the outer minimization seeks parameters at flat (robust) minima.

To stabilize sharpness across domains, we (i) use a refiner network (see Section 3.4) to produce soft labels $\tilde{y}_{ui}$, and (ii) compute a soft-label binary cross-entropy loss:

$$\tilde{y}_{ui} = f(\mathbf{h}_{u,i}; \phi), \quad (15)$$

$$\mathcal{L}_{\text{SAM}}(\theta, \phi) = -\frac{1}{|\mathcal{D}|} \sum_{(u,i) \in \mathcal{D}} [\tilde{y}_{ui} \log \hat{y}_{ui} + (1 - \tilde{y}_{ui}) \log(1 - \hat{y}_{ui})], \quad (16)$$

where $\tilde{y}_{ui}$ is the soft label and $\hat{y}_{ui}$ the model's prediction.

Following a first-order Taylor expansion (for small ρ), the optimal perturbation is estimated as:

$$\epsilon = \rho \frac{\nabla_{\theta} \mathcal{L}_{\text{SAM}}(\theta, \phi)}{\|\nabla_{\theta} \mathcal{L}_{\text{SAM}}(\theta, \phi)\|_2}. \quad (17)$$

Unlike conventional SAM [8], we apply perturbations only to the top- k parameters (by magnitude) of the semantic ID embedding fusion MLP (θ_m) and the cross-domain transfer network (θ_t). We denote the concatenated parameter vector as $\theta_{t,m} = [\theta_t; \theta_m]$ and compute:

$$\epsilon_{t,m} = \text{TopKMask} \left(\rho \frac{\nabla_{\theta_{t,m}} \mathcal{L}_{\text{SAM}}(\theta, \phi)}{\|\nabla_{\theta_{t,m}} \mathcal{L}_{\text{SAM}}(\theta, \phi)\|_2} \right), \quad (18)$$

where $\text{TopKMask}(\cdot)$ retains only the top k elements (by absolute magnitude), zeroing the rest. We γ to control the top k sparsity ratios.

The perturbed parameters ($\theta + \epsilon_{t,m}$) are then used to compute the SAM loss:

$$\mathcal{L}(\theta + \epsilon_{t,m}, \phi) = -\frac{1}{|\mathcal{D}|} \sum_{(u,i) \in \mathcal{D}} [y_{ui} \log \hat{y}'_{ui} + (1 - y_{ui}) \log(1 - \hat{y}'_{ui})], \quad (19)$$

where $\hat{y}'_{ui}$ denotes the prediction under the perturbed parameters ($\theta + \epsilon_{t,m}$). y_{ui} is the ground truth label.

Notably, during the entire SAM optimization process, the refiner parameters ϕ are held fixed. All SAM perturbations, gradient computations, and parameter updates are performed exclusively with respect to the backbone parameters θ . This ensures that the refiner consistently outputs soft labels and is not affected by sharpness-aware perturbations or updates.

3.4 Sharpness-Consistent Refining

Standard sharpness-aware minimization (SAM) computes perturbations based on the gradient of the aggregated training loss across all samples. In cross-domain scenarios, the sample size of

Algorithm 1 SC²Rec framework

```

1: while not converged do
2:   Sample batch  $\mathcal{D}$  of  $(u, i)$  from warm/cold domains.
3:   For each  $(u, i) \in \mathcal{D}$ , compute  $\mathbf{h}_{u,i}$  and predictions  $\hat{y}_{ui}$ .
4:   Generate soft label  $\tilde{y}_{ui} = f(\mathbf{h}_{u,i}; \phi)$ . (fixed  $\phi$ )
5:   Compute SAM loss  $\mathcal{L}_{\text{SAM}}(\theta, \phi)$ 
6:   Compute perturbation  $\epsilon_{t,m}$ 
7:   Compute perturbed loss  $\mathcal{L}(\theta + \epsilon_{t,m})$ 
8:   Update  $\theta$  ( $\phi$  fixed)
9:   Compute refiner loss  $\mathcal{L}_{\text{Refiner}}$ .
10:  Update  $\phi$  ( $\theta$  fixed)
11: end while

```

the warm domain $\mathcal{D}_{\text{warm}}$ greatly exceeds that of the cold domain $\mathcal{D}_{\text{cold}}$, causing the overall perturbation direction to be dominated by warm domain samples. Due to the distributional differences between domains, their respective loss landscapes often do not exhibit similar geometries [15]. As a consequence, conventional SAM may yield minima that are locally flat primarily for the warm domain, while remaining sharp for the cold domain—undermining cold domain generalization.

To address this issue, inspired by [15], we introduce a **Refiner** module which generates soft labels to help bridge the sharpness gap between warm and cold domains, and ensures that both domains can achieve locally flat minima. Specifically, the Refiner aligns the sharpness properties of both domains and produces soft labels $\tilde{y}_{ui}$ that faithfully reflect the true label distribution.

The domain-specific loss function $\mathcal{L}_d$ for domain $d \in \{\text{warm, cold}\}$ can be computed according to Equations 16 to 19, where the summation is taken over the corresponding domain's data subset.

Formally, we define sharpness in each domain as:

$$S_d(\theta, \phi) = \max_{\|\epsilon\|_2 \leq \rho} \mathcal{L}_d(\theta + \epsilon) - \mathcal{L}_d(\theta), \quad (20)$$

where ρ bounds the perturbation, and $\mathcal{L}_d(\cdot)$ is the SAM-based domain loss as in Equations 16–19. The sharpness inconsistency between domains is also penalized:

$$\Delta S(\theta, \phi) = |S_{\text{warm}, \phi}(\theta) - S_{\text{cold}}(\theta, \phi)|, \quad (21)$$

encouraging alignment of sharpness characteristics between domains.

In addition, to ensure that the Refiner's soft labels are consistent with the true binary label distribution for each user-item pair $(u, i) \in \mathcal{D}$, we impose a binary cross-entropy loss:

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{|\mathcal{D}|} \sum_{(u,i) \in \mathcal{D}} [y_{ui} \log \tilde{y}_{ui} + (1 - y_{ui}) \log(1 - \tilde{y}_{ui})], \quad (22)$$

where y_{ui} is the true label and $\tilde{y}_{ui}$ is the Refiner's soft label output.

To explicitly promote flat minima in the warm domain, we incorporate only $S_{\text{cold}}(\theta, \phi)$ as a regularization term in the Refiner loss. Thus, the overall objective for the Refiner integrates: (i) soft label accuracy via binary cross-entropy, (ii) sharpness consistency across domains, and (iii) explicit cold-domain flatness regularization:

$$\mathcal{L}_{\text{Refiner}} = S_{\text{cold}}(\theta, \phi) + \lambda \Delta S(\theta, \phi) + \mathcal{L}_{\text{BCE}}, \quad (23)$$

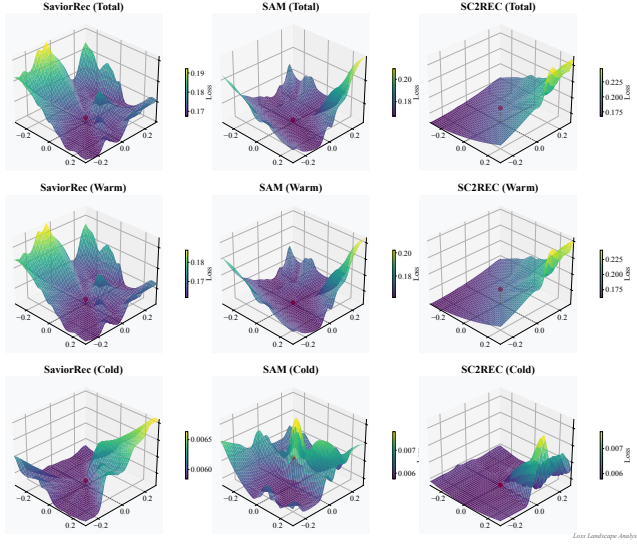


Figure 3: Loss landscape visualization. We adopt the visualization technique from Li et al. [16], which perturbs model parameters along two randomly selected directions to generate a three-dimensional loss surface.

where λ is balancing ratio, and θ and ϕ denote the backbone and Refiner parameters, respectively. Here, $\Delta S(\theta, \phi)$ encourages alignment of loss landscape sharpness between the warm and cold domains.

During Refiner training, we update only the Refiner parameters ϕ while keeping the backbone parameters θ fixed. This scheme ensures that the Refiner consistently guides the warm domain toward flat minima—thus enhancing generalization—while maintaining high fidelity between the generated soft labels and the true labels. The complete training procedure is detailed in Algorithm 1.

4 Experiment

4.1 Dataset

To address the critical scarcity of realistic, semantic ID enriched datasets for cold-start industrial recommendation scenarios [37], we publicly release a large-scale benchmark dataset derived from one of the world’s largest social media platforms. Our benchmark focuses on recommending truly new items—specifically, content such as articles and images with fewer than 30 page views, a threshold adopted from platform production practices to indicate cold-start status.

The dataset is temporally partitioned, using the interactions from three consecutive days for training and the following day for testing. For public accessibility and privacy, 1% of daily data instances are sampled; stratified sampling is performed to preserve the original distribution of key attributes, and results are reported for both the sampled and complete data. We construct both a warm domain (frequent user-item interactions) and a cold domain (new items with fewer than 30 page views), each drawn from the same platform. A key feature of our dataset is the provision of semantic ID codes for all items and user behavioral sequences. These codes are

Table 2: Dataset statistics.

Dataset	Industrial (1%)		Industrial (Full)	
Domains	Warm	Cold	Warm	Cold
Impressions	8.93M	0.41M	900M	42M
Clicks	0.36M	10.8K	36M	1.1M

generated in two stages: (1) large language models (LLMs), such as CLIP and GPT, which are finetuned on our dataset to produce multimodal item embeddings, and (2) a Residual Quantized Variational Autoencoder (RQ-VAE), which is subsequently trained to generate compact semantic IDs and sequence codes. By offering real-world cold-start splits, comprehensive semantic ID codes, and 500+ sparse and dense features, our dataset enables new directions in semantic ID-based, cross-domain, and cold-start recommendation research. Table 2 shows the data statistics.

4.2 Baselines

We compare our proposed method against nine representative baselines encompassing parameter sharing, transfer learning, and attention-based approaches, all pertinent to cross-domain and cold-start recommendation. For a fair comparison within our semantic ID-based recommendation framework, we adapt each baseline to utilize semantic ID representations as input features. This unified protocol enables systematic assessment of each cross-domain method’s efficiency under consistent conditions.

- **Shared Bottom (Shbot)** [1]. Shbot employs a shared feature extraction layer across all domains and tasks, facilitating knowledge transfer. Independent task-specific prediction towers are subsequently applied to each domain.
- **MMOE** [20]. MMOE incorporates multiple shared expert subnetworks with task-specific gating mechanisms that dynamically select experts for each domain, enhancing cross-domain transfer while mitigating negative transfer.
- **STAR** [25]. STAR combines shared networks for extracting general domain information with domain-specific subnetworks that use partition normalization to capture distinctive features.
- **SARNet** [24]. SARNet integrates scenario-specific and scenario-shared expert layers, augmented by an attention mechanism, to enable effective cross-domain knowledge transfer and adaptive feature extraction.
- **Hamur** [18]. Hamur introduces a weight generator and adapter module to inject domain information into the intermediate layers of prediction towers.
- **DAA** [27]. DAA employs dual discriminators and adversarial objectives to minimize domain shift and alleviate negative transfer in cross-domain scenarios.
- **Diff-MSR** [30]. Diff-MSR leverages diffusion-based representation generation to extract useful features from the warm domain, which are then transferred to improve recommendations in the cold domain.

Table 3: Ablation study results of SC²Rec on the Industrial (1% for release) and Industrial (Full) datasets. We report the CTR AUC and Logloss (LL). Bold values with “*” indicate statistically significant improvements over the ablation variants, as determined by a two-sided t-test ($p < 0.05$).

Dataset	Industrial (1%)				Industrial (Full)			
Backbone	Warm,AUC	Cold,AUC	Warm,LL	Cold,LL	Warm,AUC	Cold,AUC	Warm,LL	Cold,LL
SC ² Rec w/o all	0.6793	0.6949	0.1627	0.1233	0.7925	0.8200	0.1712	0.1328
SC ² Rec w/o Refiner	0.6791	0.7001	0.1629	0.1220	0.7923	0.8244	0.1714	0.1319
SC²Rec	0.6827*	0.7105*	0.1621*	0.1214*	0.7959*	0.8299*	0.1711*	0.1304*

- **SaviorRec** [33]. SaviorRec introduces a semantic ID codebook within the ranking module for improved alignment between semantic IDs and utilizes a Bi-directional Target Attention Mechanism to fuse information from both semantic and unique ID sources.
- **SCDR** [35]. SCDR employ standard SAM to improve the generalization of user embedding. For fair comparison, we enhance SCDR and employ standard SAM to sharing parater-meters in Shbot.

4.3 Experiment Setups

We evaluate ranking performance and prediction error using the Area Under the Receiver Operating Characteristic Curve (AUC) [12] and Logloss [38]. Each model is trained and tested over five independent runs with randomly chosen seeds; for each metric, we report the mean across all runs. Statistical significance between models is assessed using paired t -tests. Baseline implementations adhere to protocols from their respective original papers, with corrections and enhancements applied as necessary. For the sampled dataset, all embedding vectors are set to dimension 32, while those for the full dataset use dimension 64. Neural predictors for the warm and cold domains are implemented as multilayer perceptrons (MLPs) with layer sizes [128, 64, 1]. Expert networks are constructed as MLPs with a single layer of size [512] for the sampled data and two layers [512, 256] for the full dataset. We utilize three residual quantization layers with progressively increasing codebook sizes: $(K_1, K_2, K_3) = (1,000, 60,000, 1,200,000)$. All models are trained with a batch size of 8,096 and optimized using Adam with a learning rate of 1×10^{-3} . Experiments are conducted using PyTorch 2.0 in deterministic mode on NVIDIA A10 GPUs.

4.4 Overall Performance Comparison

Table 1 presents a comprehensive comparison of SC²Rec with nine competitive baselines on both the released 1% industrial datasets and the full-scale industrial datasets. On the 1% released dataset, SC²Rec consistently outperforms all baselines, achieving an AUC of 0.6807/0.7105 and a LogLoss of 0.1621/0.1214 for the warm and cold domains, respectively. Notably, in the cold-start setting, SC²Rec surpasses the second-best method, SCDR, by 1.51% in AUC and by 0.0006 in LogLoss, with all improvements being statistically significant ($p < 0.05$). A similar pattern is observed for the full dataset, where SC²Rec attains AUCs of 0.7949/0.8299 and LogLoss values

of 0.1711/0.1304 in the warm and cold domains, respectively, outperforming SCDR by 0.61% in AUC and 0.0006 in LogLoss. Given the industrial-scale feature space (over 500 dimensions), such performance improvements are notable, even if absolute gains tend to be modest in full-scale settings—a common phenomenon in industrial recommendation due to extensive feature diversity and sample abundance.

In the warm domain, the performance gap between advanced methods such as Shbot, MMOE, SARNet, and Hamur is comparatively small. This is largely attributable to the abundance of user interactions and key semantic IDs, which allow most models to effectively capture and generalize user preferences. As a result, the benefits of complex architectures or sharpness-aware optimization are less pronounced, since the available data sufficiently supports robust representation learning. Although SC²Rec still achieves the strongest results, its margin of improvement over baselines is naturally reduced relative to the cold domain, owing to better generalization facilitated by sharpness-aware minimization.

In contrast, the cold domain poses substantial challenges due to data sparsity. Baselines like Shbot and MMOE leverage semantic IDs for knowledge transfer, yielding performance similar to SaviorRec. SCDR further incorporates sharpness-aware minimization (SAM), producing the second-best results in all metrics. Building on this foundation, SC²Rec introduces a Refiner module designed to explicitly align the loss landscapes of the warm and cold domains, enforcing greater consistency in loss sharpness. This approach mitigates shortcomings of standard SAM-based methods, wherein the loss landscape is dominated by warm domain samples and sharpness optimization for the cold domain remains insufficient. Consequently, SC²Rec achieves significantly higher AUC and lower LogLoss in cold-start scenarios, thus validating the effectiveness of our sharpness-consistent refinement strategy.

In terms of scalability and computational efficiency, although SC²Rec results in a moderate increase in training time, we adopt the parallel gradient computation strategy from [32] to effectively mitigate this overhead. Consequently, SC²Rec only incurs moderate additional resource requirements (5% increase in training time and memory usage), which is entirely acceptable for training cost in practical scenarios. Importantly, since the sharpness-consistent objectives and refiner network are exclusively applied during training, our framework does not impose any extra cost at inference time—a crucial consideration for the scalability of recommender systems in real-world applications.

4.5 Visualization

To further analyze the optimization dynamics of different cross-domain recommendation methods, we visualize the 3D loss landscapes of SaviorRec, SCDR, and our proposed SC²Rec in both the warm and cold domains (see Figure 3). Following the approach of Li et al. [16], we perturb model parameters along two randomly selected directions and evaluate loss values on the test dataset, thus constructing 3D loss surfaces from the high-dimensional parameter space.

SaviorRec (left column) presents a sharply curved loss landscape across all domains (average curvature: 0.0166 in the overall model, 0.0164 in the warm domain, and 0.0013 in the cold domain). Reflecting this, Table 1 shows SaviorRec’s performance is suboptimal, especially under cold-start conditions (1% data: AUC 0.6948, LogLoss 0.1226), indicating more pronounced generalization bottlenecks associated with sharp minima.

In contrast, SCDR (middle column) applies Sharpness-Aware Minimization (SAM), yielding a flatter loss landscape in the warm domain (curvature: 0.0087 overall, 0.0085 warm). However, the cold domain landscape remains relatively steep (curvature: 0.0015), likely due to sample imbalance that prioritizes warm domain optimization. Curvature is estimated via the second derivative at the parameter space center. Correspondingly, SCDR achieves strong but not optimal cold-start performance (AUC 0.6999, LogLoss 0.1220), the second best among all baselines.

SC²Rec (right column), equipped with the Refiner module, attains the flattest and most consistent loss surfaces (curvature: 0.0054 for both overall and warm domains, and a notably low 0.0004 in the cold domain). This sharpness consistency is closely reflected in its superior quantitative metrics (AUC 0.7105, LogLoss 0.1214 on the cold domain of the 1% dataset), as reported in Table 1, where SC²Rec achieves the best results across all metrics and domains. Notably, the visualization corroborates these empirical findings: smoother loss landscapes correspond to improved generalization and elevated performance, especially under cold-start settings, thus validating the impact of SC²Rec’s sharpness-consistent refinement approach.

4.6 Ablation Study

We evaluate the impact of each SC²Rec component via ablation studies on Industrial (1%) and Industrial (Full) datasets. We compare the full model to (i) *w/o all* (removing Sharpness-Aware Minimization and Refiner) and (ii) *w/o Refiner* (removing only the Refiner).

As shown in Table 3, the full SC²Rec consistently outperforms both variants across all metrics. On Industrial (1%), it achieves a cold-domain AUC/LogLoss of 0.7105/0.1214, versus 0.6949/0.1233 (*w/o all*) and 0.7001/0.1220 (*w/o Refiner*). Similar significant gains ($p < 0.05$) are observed on the full dataset. Results show that Sharpness-Aware Minimization substantially improves performance, while the Refiner provides additional benefits—especially for cold-start. Both Sharpness-Aware Minimization and Refiner are essential for SC²Rec’s superior generalization.

4.7 Hyper-Parameter Sensitivity Analysis

We evaluate SC²Rec’s sensitivity to key hyper-parameters on the industrial dataset, using both 1% and full training splits (Figure 4).

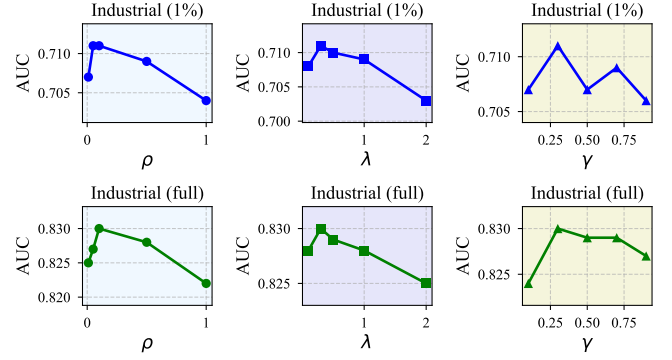


Figure 4: Hyper-parameter sensitivity analysis of SC²Rec. AUC performance is reported on the industrial dataset with 1% and full training data. We vary (a) the perturbation ratio ρ , (b) sharpness inconsistency loss ratio λ , and (c) the sparsity ratio γ .

We report AUC while varying the perturbation ratio (ρ), sharpness inconsistency loss ratio (λ), and sparsity ratio (γ).

Perturbation ratio ρ . AUC is relatively stable as ρ varies, peaking for moderate perturbation levels $\rho \in [0.05, 0.1]$ (AUC: 0.711 for 1%, 0.827–0.830 for full). Larger ρ slightly degrades performance, indicating that excessive perturbation introduces harmful noise. We therefore set $\rho = 0.1$ in subsequent experiments.

Sharpness inconsistency loss ratio λ . Performance improves as λ increases up to 0.3–0.5 (AUC: 0.711 for 1%, 0.830 for full), with higher λ leading to marginal declines. We set $\lambda = 0.5$ for final model training.

Sparsity ratio γ . AUC is maximized for $\gamma = 0.3$ –0.7 (0.711 for 1%, 0.829–0.830 for full), while overly large γ hinders performance. We choose $\gamma = 0.7$.

5 Conclusion

In this work, we addressed the persistent cold-start challenge in recommendation systems by introducing SC²Rec, a novel sharpness-consistent cross-domain recommendation framework. By applying sharpness-aware minimization to transfer parameters and incorporating a Refiner network that enforces landscape flatness in both domains, SC²Rec effectively fosters balanced generalization and improved cross-domain transfer. Extensive experiments on industrial datasets demonstrate that SC²Rec significantly outperforms existing state-of-the-art approaches in cold-start scenarios. To further advance research in this field, we also released a large-scale, real-world cold-start benchmark with truly unseen items and rich semantic information.

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